Linear Algebra II HW 8

Due Date: on or before 18:00, Apr/24/2025 (Thursday), in tutorial.

1. Let $A=\begin{pmatrix}\alpha+\beta & -\alpha\beta \\ 1 & 0\end{pmatrix}$ where α and β are distinct real numbers. Let I be the 2×2 identity matrix.

   (a) Show that $A^2=(\alpha+\beta)A-\alpha\beta I$.

   (b) Using (a), or otherwise, show that $(A-\alpha I)^2=(\beta-\alpha)(A-\alpha I)$ and $(A-\beta I)^2=(\alpha-\beta)(A-\beta I)$.

   (c) Let $X=s(A-\alpha I)$ and $Y=t(A-\beta I)$ where s and t are real numbers.
   Suppose $A=X+Y$.

   (i) Find s and t in terms of α and β .

   (ii) For any positive integer n , prove that

   $$X^n=\frac{\beta^n}{\beta-\alpha}(A-\alpha I) \quad \text{and} \quad Y^n=\frac{\alpha^n}{\alpha-\beta}(A-\beta I) .$$

   (iii) For any positive integer n , express A^n in the form of $pA+qI$, where p and q are real numbers.
   [Note: It is known that for any 2×2 matrices H and K ,
   if $HK=KH=\begin{pmatrix}0 & 0\\ 0 & 0\end{pmatrix}$, then $(H+K)^n=H^n+K^n$ for any positive integer n .]

2. Find an orthogonal basis for the column space of $A=\begin{pmatrix}1 & 3 & 5\\ 1 & 1 & 0\\ 1 & 1 & 2\\ 1 & 3 & 3\end{pmatrix}$. Find a QR factorization of A. Hence find the least-squares solution of the inconsistent system $A\boldsymbol{x}=\boldsymbol{b}$ where $\boldsymbol{b}=(1,2,3,4)^T$.

Find the eigenvalues and eigenvectors of the matrices.

3. $A = \begin{pmatrix} 3 & -1 \\ 4 & -2 \end{pmatrix}$.

4. $A = \begin{pmatrix} 1 & 0 \\ 6 & -1 \end{pmatrix}$.

5. $A = \begin{pmatrix} -1 & 4 & -2 \\ -3 & 4 & 0 \\ -3 & 1 & 3 \end{pmatrix}$.